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CROSS-VALIDATION FOR METAMODELS

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2026-09-08

Motivations and Outline

- ▶ Simulations can be computationally expensive. We would like to find ways to make accurate approximations of their results in shorter amounts of time
- ▶ The model's performance on training data might be a poor indicator of how the model will perform in a real world scenario. We want to to simulate a scenario with unseen data.

Talk structure

1. **Basics:** metamodels, validation, and cross-validation
2. **CV schemes:** k -fold and leave-one-out
3. **Case study:** CO₂ WAG surrogate validation

What is a metamodel?

- ▶ “Model of a model”
- ▶ Focus on input-output behaviour of complex systems or models
- ▶ Used to reduce time, cost, computational effort etc.

Our example

Instead of rerunning OPM Flow for every combination of CO₂ and water injection rates, a metamodel predicts retained CO₂ directly.

Validation vs. verification

Validation

- ▶ Is the model behaving correctly
- ▶ Does the model generalize well to unseen data
- ▶ Is the model's performance good enough

Verification

- ▶ Is the intended model implemented correctly
- ▶ Are specifications met
- ▶ Is the correct training data being used

Why do cross-validation?

- ▶ Quantify estimator performance
- ▶ Remove validation score's dependency on the data split
- ▶ Check model's ability to generalize
- ▶ Guide model selection

Two approaches to cross-validation

Leave- p -out

- ▶ Use p values for validation, rest for training
- ▶ Repeat with different training and validation cuts until all possible cut sets are tested

k -fold

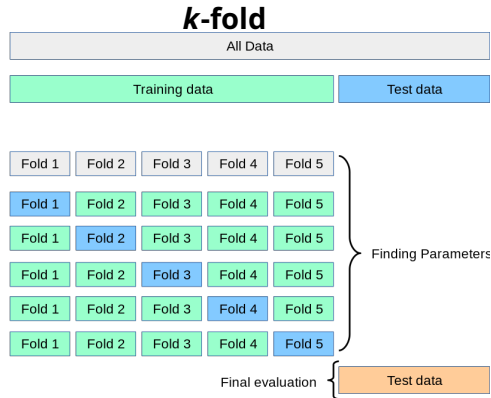
- ▶ Partition data into k subsets of equal size (aka. folds)
- ▶ One fold is used for validation, the rest for training
- ▶ Repeat k times, using each fold for validation exactly once
- ▶ If $k = n$ this is the same as leave-one-out

Two approaches to cross-validation



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<https://commons.wikimedia.org/w/index.php?curid=87684543>



Credits: scikit-learn

Practical considerations for metamodel cross-validation

- ▶ DoE: ensure broad coverage of input domain to avoid extrapolation issues
- ▶ Size of training data: large datasets can make CV slow, Leave-p-out takes much longer than k-fold CV
- ▶ Avoid data leakage between training, validation, and test sets
- ▶ Use reasonable error metrics and remember normalization

Model	Scheme	Fits	Time [s]	RMSE	Q^2
RSM	5-fold	5	0.009	57094	0.9877
RSM	LOOCV	300	0.274	57170	0.9877
GBR	5-fold	5	0.257	49940	0.9906
GBR	LOOCV	300	16.85	45672	0.9921

LOOCV costs many more fits; here it gives a similar conclusion.

CO₂ WAG case study

We use two operating controls:

$$\mathbf{x} = [q_{\text{gas}}, q_{\text{water}}]$$

Response:

$$y = M_{\text{CO}_2, \text{retained}}$$

- ▶ 300 Latin Hypercube runs
- ▶ 300 random-design runs



Metamodel predictions are near-instant

Take-home message

When simulations and computations are hard, metamodels can replace repeated full-physics runs with near-instant predictions.

Model	Train [s]	Predict [ms]	Speed-up
Quadratic RSM	0.003	0.13	2.3×10^4
GBR	0.054	0.10	3.0×10^4
Gaussian process	0.577	0.13	2.4×10^4

Simulator scale

$$t_{\text{OPM}} \approx 3 \text{ s/run}$$

Metamodel scale

$$t_{\text{pred}} \approx 0.1 \text{ ms/run}$$

Training fit is not predictive validation

Take-home message

Models may look good during training but break on excluded simulator runs; validation must test out-of-fold predictions.

Training error

Evaluate on rows used for fitting.

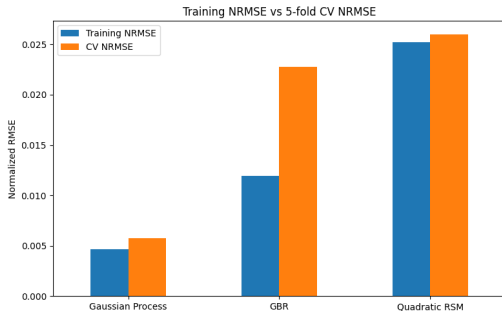
$$\hat{y}_i^{\text{train}} = \hat{f}(\mathbf{x}_i)$$

CV error

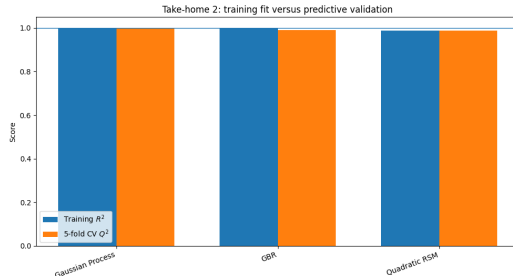
Evaluate out-of-fold predictions.

$$\hat{y}_i^{\text{CV}} \text{ from a model excluding } i$$

A good training fit does not guarantee predictive accuracy



NRMSE



Q^2

Take-away

A low training error alone can be misleading.

Validation with unseen data is needed to assess generalization.

Repeated CV checks whether the split was unusually lucky



Model	Mean Q^2	SD Q^2
Gaussian process	0.9994	0.0001
GBR	0.9907	0.0025
Quadratic RSM	0.9874	0.0025

Point

A single random split can be lucky or unlucky; repeated CV shows the stability of the validation score.

The split and the DoE define the claim

Take-home message

The validation split defines what “generalization” means, while the DoE controls where predictions are supported by simulator runs.

Random 5-fold asks

Can we predict new points interspersed among sampled operating points?

Quadrant holdout asks

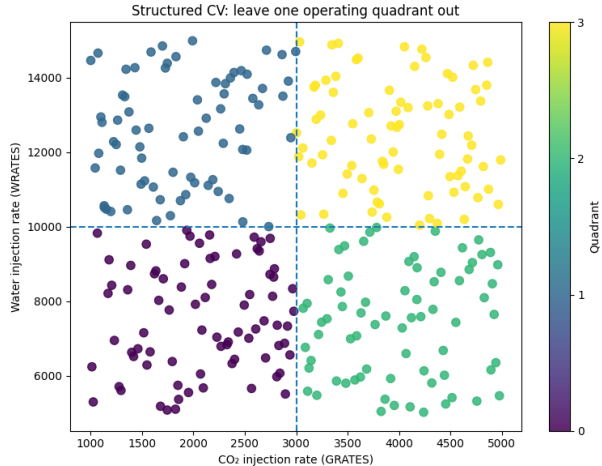
Can we predict a whole operating region absent from fitting?

Structured holdout removes a whole operating region

$$q_{\text{gas}} = 3000, \quad q_{\text{water}} = 10000$$

The domain is split into four operating quadrants.

One quadrant is held out at a time.

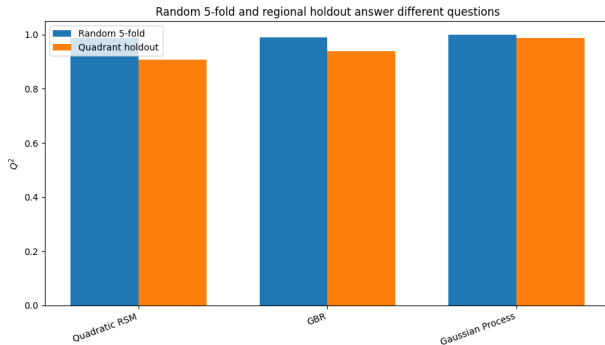


The same data can give a different validation claim

Random 5-fold mainly tests interpolation.

Quadrant holdout tests prediction with weaker local support.

The score changes because the question changes.

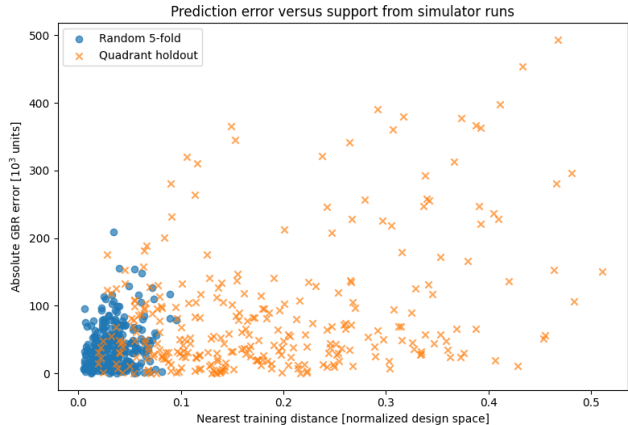


Poor geometric support increases error risk

For each validation point:

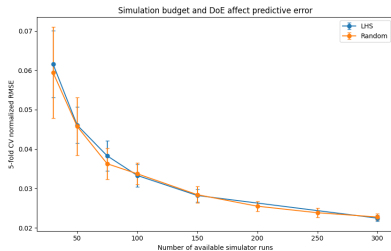
$$d_i = \min_{\mathbf{x}_j \in D_{\text{train}}} \|\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j\|_2$$

Large d_i means weaker support from nearby simulator runs.

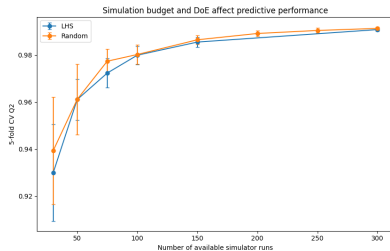


Simulation budget and DoE affect CV reliability

Learning curves using GBR: as more simulator runs become available, predictive error falls, Q^2 moves toward 1, and the difference between LHS and random sampling becomes smaller.



- ▶ NRMSE decreases as N grows.
- ▶ Q^2 increases toward 1.



- ▶ Variability decreases with N .
- ▶ LHS and random become more similar for large N .

Summary: Three take-home messages

1. Simulations and computations may be hard; metamodels make repeated evaluations practical.
2. Models may look good during training but break; predictive validation must use excluded simulator runs.
3. CV scores are conditional on the DoE and the validation split.

CV is not just a number; it is a claim about where the metamodel is trusted.

References and data

- ▶ Lligüizaca-Dávila et al. (2026), *CO₂ WAG and SAG machine learning data-driven models in an open-source framework*.
- ▶ Public data repository: <https://github.com/Silver9232/co2-wasp>
- ▶ Stone (1974), cross-validatory choice and assessment of statistical predictions.
- ▶ McKay, Beckman and Conover (1979), Latin hypercube sampling.
- ▶ Rasmussen and Williams (2006), *Gaussian Processes for Machine Learning*.